Dataset-Routed Clustering with Candidate Pair Constraints for AnimalCLEF 2026

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Abstract

AnimalCLEF 2026 evaluates individual animal discovery and re-identification as an open-set clustering task across four dataset namespaces: `LynxID2025`, `SalamanderID2025`, `SeaTurtleID2022`, and `TexasHornedLizards`. Each test image must be assigned to an identity cluster and submissions are evaluated using Adjusted Rand Index. We describe the VIPL-VSU submission, a dataset-routed clustering pipeline that combines foundation wildlife descriptors, supervised metric-learning heads, local feature evidence, learned pairwise fusion, transductive smoothing, and candidate pair constraints. The core idea is to decompose clustering into two stages: a high-recall automatic stage that retrieves top-K candidate links, followed by a precision-oriented pair-constraint stage that turns ambiguous candidates into graph constraints. This strategy was first developed for `TexasHornedLizards`, where the official Texas horned lizard study motivated ventral-pattern preprocessing and local matching, and was later reused for `SalamanderID2025`. Our final submission obtained 0.67987 public ARI and 0.69632 private ARI, ranking third on the private leaderboard.

Keywords

animal re-identification, open-set clustering, metric learning

1. Introduction

Individual animal re-identification aims to recognise the same animal across photographs collected at different times, locations, viewpoints, and acquisition conditions. It is an important tool for ecological monitoring because many species can be identified from persistent visual markings such as coat patterns, scales, scutes, stripes, spots, or ventral markings. Compared with human re-identification, animal re-identification is often more open-ended: the number of individuals is not fixed, many individuals have only a few images, backgrounds and camera traps can be highly biased, and the visual cue differs substantially across species [1, 2].

AnimalCLEF 2026, organised within LifeCLEF 2026, casts this general problem as discovery-oriented clustering across four animal datasets [3, 4]. This setting makes the objective different from ordinary retrieval: a model with high nearest-neighbour recall can still fail if its clustering threshold merges different individuals, while a conservative pipeline can fail by over-splitting the same animal. The official metric is Adjusted Rand Index (ARI), a chance-corrected measure of agreement between two partitions [5].

Our final pipeline was built around one observation: the four datasets expose different visual and statistical structures, so a single backbone and threshold is not robust. Lynx images contain relatively stable coat patterns and strong cluster-count priors. Salamanders require fine local texture and careful ambiguity handling. Sea turtles are already well served by frozen wildlife descriptors. Texas horned lizards have no labelled local train split in the released package, making them a strong domain-shift and self-training problem. We therefore use a dataset-routed pipeline in which each species branch has its own representation, pair evidence, thresholding, and post-processing policy.

The key simplification is to avoid treating clustering as one monolithic thresholding problem. We instead split it into a high-recall first stage and a precision-oriented second stage. The first

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stage is allowed to be inclusive: global descriptors, local ORB evidence, and species-specific preprocessing only need to surface plausible same-individual links. The second stage is selective: candidate pair verification converts a much smaller ambiguity set into positive and negative graph constraints.

The verification component of our pipeline was inspired by ambiguity-aware pair sampling for animal re-identification [6]. That work frames feedback as pairwise constraints, producing must-link and cannot-link signals for constrained clustering. We adopt the same pairwise interface in a narrower engineering form: the model proposes candidate pairs, and the verification stage only accepts or rejects pair edges. In our competition setting, these decisions are compiled into clustering constraints over top-K local matches and uncertain clusters.

Our contributions are:

- A practical dataset-routed clustering pipeline for AnimalCLEF 2026, with official scores of 0.67987 public ARI and 0.69632 private ARI.
- A two-stage clustering decomposition that uses automatic top-K candidate retrieval for recall and pair-constraint verification for precision.
- An account of the species-specific routes that proved useful: transductive seed smoothing for lynx, learned pairwise fusion and graph constraints for salamanders, conservative descriptor fusion for sea turtles, and trusted-view self-training for Texas horned lizards.
- Documented intervention evidence showing that many useful components were not universal: ORB local matching helped salamanders but was not used for lynx or sea turtles, yellow-pattern cues hurt as hard vetoes but helped as learned XGBoost features, and Texas descriptor concatenation degraded despite preserving seed pairs.

2. Related Work

2.1. Animal Re-Identification

Animal re-identification has historically relied on species-specific visual cues and semi-automatic photo-identification tools, especially for animals with stable markings [1]. Recent benchmarks and toolkits have made the setting more comparable across species. WildlifeDatasets collects many individual animal datasets and provides a common evaluation interface, and MegaDescriptor is one of the strongest general descriptors from this line of work [2]. MiewID is another multispecies model trained from community-curated identity data and was complementary to MegaDescriptor in our experiments [7]. Dataset-specific benchmarks are also important: SeaTurtleID2022 emphasizes long-span open-set re-identification under realistic temporal splits [8], while CzechLynx provides a large camera-trap resource for Eurasian lynx individual identification and related vision tasks [9].

2.2. Local Matching and Species-Specific Cues

Local feature matching is useful when identity is encoded in spatially localized markings. ORB provides a fast binary keypoint descriptor that can be paired with geometric verification [10], while modern matchers such as LightGlue offer stronger learned correspondences for image matching [11]. For Texas horned lizards, the official reference study by Biffi et al. showed that ventral spot patterns carry individual identity information: HotSpotter placed the correct previous image first for 85 of 90 known recapture photos, and in four additional cases the correct match was still among the top-10 recommendations [12]. This supports a useful separation between local matching as candidate retrieval and later decisions about which candidate links are reliable.

Table 1
Dataset sizes in the released AnimalCLEF 2026 package used by our pipeline.

Dataset	Train images	Test images	Main visual cue / source
LynxID2025	2,957	946	coat/flank markings; CzechLynx-related [9]
SalamanderID2025	1,388	689	dorsal yellow spot patterns [13]
SeaTurtleID2022	8,729	500	facial and scute patterns [8]
TexasHornedLizards	0	274	ventral/body spot patterns [12]
Total	13,074	2,409	–

2.3. Constrained Clustering and Pair Feedback

Open-set re-identification pipelines often need to convert retrieval evidence into a partition, so pair-level decisions become useful when descriptor scores alone are ambiguous. Ambiguity-aware pair sampling for animal re-identification converts selected pairwise feedback into must-link and cannot-link constraints for clustering [6]. We use this line of work as the conceptual basis for treating verified candidate pairs as graph constraints.

3. Task and Data

3.1. Task Definition

The goal of AnimalCLEF 2026 is to discover individual animals in the hidden test split and submit the resulting identity partition. In other words, participants are not asked to predict species labels or to assign images to a fixed set of known training identities. They must group the unlabelled test images so that photographs of the same animal receive the same cluster identifier and photographs of different animals receive different identifiers.

The input consists of four dataset namespaces, labelled training identities where available, and unlabelled test images. The required submission is a CSV file with two columns, `image_id` and `cluster`. The official evaluation compares the submitted partition with the hidden ground-truth identity partition using Adjusted Rand Index (ARI). We therefore treat AnimalCLEF 2026 as an open-set clustering task, where retrieval scores are only intermediate evidence and the final object being evaluated is the complete test-set clustering.

3.2. Dataset Splits

The officially released data contain 15,483 images in four dataset namespaces. The train split contains 13,074 labelled images for three datasets, while `TexasHornedLizards` is provided as a test-only dataset with 274 images. The test split contains 2,409 images in total. The datasets are released through the AnimalCLEF benchmark [4]; where available, we also cite the underlying dataset or species study used by the benchmark [13, 8, 9, 12].

The labelled train distribution is strongly long-tailed. Across the three labelled datasets there are 1,102 identities and 317 singleton identities. Salamander is the most extreme case: `SalamanderID2025` has 587 train identities, of which about 52.8% are singletons. This matters because identity-level validation and pairwise training produce very different support across datasets. Figure 1 shows the per-dataset identity-size distribution that motivated the dataset-routed design.

Local validation protocol. Local validation refers to our internal experiment protocol, not the official Kaggle evaluation split. For each labelled dataset, identities are shuffled with seed 42 and approximately 10% of identities are held out; all images of the same individual stay on the same side of the split. This produces 991 training-side identities with 11,739 training-side images

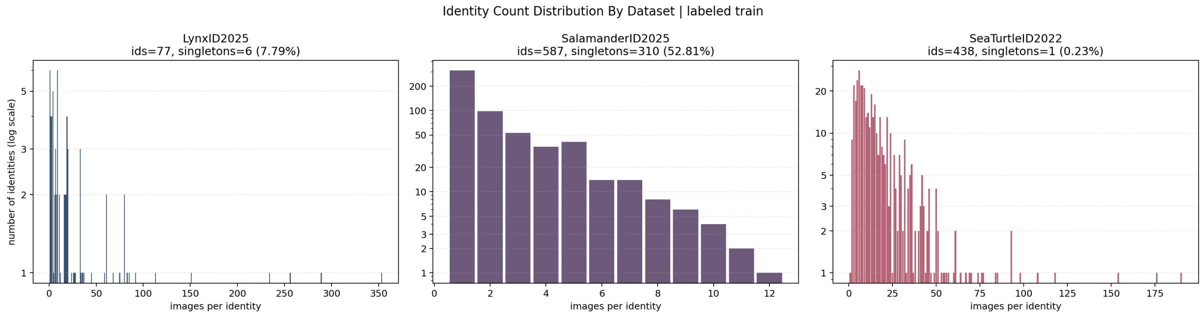

Figure 1: Identity-size distribution in the labelled training split. Salamander is dominated by singleton identities, while sea turtle has many multi-image identities and lynx has fewer identities with a long-tailed image count. This motivates identity-level validation and species-specific clustering policies.

and 111 validation identities with 1,335 validation images. The validation split contains 69/8 training/validation identities for lynx, 528/59 for salamander, and 394/44 for sea turtle. Official competition scores are reported separately from the Kaggle public and private leaderboards.

We report ARI as the main clustering metric. We also track Normalized Mutual Information, pairwise precision/recall/F1, Recall@1/5/10, cluster count, and singleton-cluster ratio. Retrieval metrics are treated only as candidate-recall diagnostics: a high Recall@10 means that the global route can supply useful candidates, but the final objective remains the ARI of the cluster partition.

4. Pipeline Overview

Figure 2 summarises the pipeline. Each image is routed by dataset. Most branches start from two pretrained wildlife descriptors: MiewID [7] and MegaDescriptor [2]. We use cosine similarity on L2-normalised embeddings, average-linkage clustering for descriptor-only routes, and graph-style post-processing where pairwise evidence is more reliable than global thresholds.

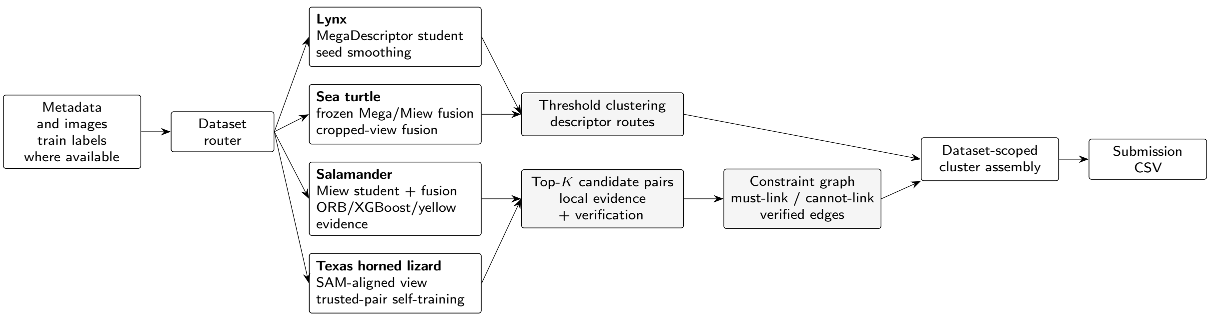

Figure 2: Pipeline overview. Images are first routed by dataset. Lynx and sea turtle use descriptor-based threshold clustering, while salamander and Texas horned lizard use high-recall top-K candidate pairs followed by pair verification and constraint-graph compilation. The dataset-scoped cluster outputs are then assembled into the submitted CSV.

For a descriptor matrix E with L2-normalised rows, the base similarity is

$$s(i, j) = E_i^{\top} E_j. \tag{1}$$

Here E_i and E_j denote the embeddings of images i and j, and $s(i, j)$ is their cosine similarity because each row of E is L2-normalised. For frozen descriptor fusion, MegaDescriptor and MiewID embeddings are concatenated and L2-normalised. For learned routes, a supervised student projects the backbone output into a 512-dimensional embedding and is clustered using a dataset-specific threshold.

Table 2
Final submission recipe by dataset.

Dataset	Main blocks	Dim.	Threshold	Constraint or post-processing policy
`LynxID2025`	MegaDescriptor-L supervised student; transductive seed smoothing	512	0.85	average-linkage thresholding
`SalamanderID2025`	MiewID student; frozen descriptor fusion; ORB and yellow-pattern features; XGBoost pair scoring	4200	0.25	top-K pair graph with reviewed must-link/cannot-link constraints
`SeaTurtleID2022`	frozen MegaDescriptor+MiewID fusion with cropped-view fusion	7376	0.70	conservative average-linkage thresholding; no pair constraints
`TexasHornedLizards`	SAM-aligned gray center-body views; trusted-view MiewID self-training	512	0.44	thresholding plus reviewed pair-registry repair

The descriptor-only clustering route uses cosine distance followed by average-linkage hierarchical clustering implemented with scikit-learn [14]. Each dataset is clustered independently, and thresholds are swept on validation where labels are available. The submitted labels are dataset-scoped cluster strings that preserve the row order and schema of the sample submission. Detailed branch configurations and operating parameters are reported in Appendix A.

Two-stage candidate pair verification. We use candidate pair verification as a narrow form of ambiguity-aware pair sampling. Here verification means a same-individual or different-individual decision for a retrieved image pair. The pipeline does not assign global identity labels by hand. It first retrieves a small candidate-pair set and then turns reliable pair decisions into graph constraints:

1. Preprocess images into a view that exposes the relevant individual markings.
2. Use descriptors and local matching to retrieve top-K high-recall candidate pairs.
3. Score or review only these candidate pairs.
4. Convert accepted decisions into must-link, cannot-link, split, or attach constraints.
5. Recompile the clustering graph using these constraints.

For a dataset with n images and K retrieved neighbours per query, this reduces the pair set from all possible pairs to an nK candidate list; in our main runs $K = 10$. The first stage is therefore recall-oriented, while the graph stage is precision-oriented.

For clarity, the final constrained submission is not a fully automatic rerun. It includes human-in-the-loop review of retrieved candidate pairs for salamanders and Texas horned lizards. The reviewer saw only candidate image pairs from the released competition data and marked them as same individual, different individual, or uncertain. Hidden labels were not available, and the reviewer did not edit final cluster identifiers directly. A deterministic graph compiler translated these pair decisions into must-link and cannot-link constraints. In the final submission, the salamander constraints contained 179 positive pairs and 4216 negative or default-negative pairs, and raised the public/private ARI from 0.62817/0.65096 to 0.67987/0.69632. The Texas registry contained 86 must-link and 1514 cannot-link rows and changed 8 images.

Table 3
Identity-level validation ARI for frozen descriptor baselines. Texas is excluded because no labelled train split is available locally.

Route	Lynx ARI	Salamander ARI	SeaTurtle ARI
Frozen MiewID	0.0864	0.1681	0.7854
Frozen MegaDescriptor	0.1027	0.1078	0.8903
Frozen MegaDescriptor + MiewID	0.2250	0.2030	0.9256

5. Representation Learning and Pair Evidence

5.1. Foundation Descriptors

Our first baselines used frozen MiewID-msv3 and MegaDescriptor-L embeddings. Both embeddings are L2-normalised, concatenated for early fusion, and normalised again. The two descriptors were complementary: MiewID was stronger than MegaDescriptor on salamanders, while MegaDescriptor was stronger on sea turtles. Their early fusion was the strongest no-training baseline on all three labelled validation datasets, as shown in Table 3.

5.2. Supervised Metric Learning

We fine-tuned MiewID- and MegaDescriptor-initialised models on the labelled train splits of lynx, salamander, and sea turtle using separate metric-learning classification heads for each labelled dataset. Across repeated stable retraining runs, Mega-initialised routes were strongest for lynx, while Miew-initialised routes were more useful for salamanders. Sea turtle did not consistently benefit from fine-tuning over frozen fusion. Additional training details are given in Appendix A.

5.3. Local Matching

For salamanders and Texas horned lizards, whole-image descriptors were not enough. We therefore used ORB keypoint matches [10] with RANSAC geometric verification [15] on top-K candidate neighbours. The local score is used as a conservative boost or pair feature rather than as a full replacement for global similarity. This gate improved salamander validation ARI from 0.2030 to 0.2940, while smoke tests did not justify enabling it for lynx or sea turtle. Exact matching parameters are listed in Appendix A.

5.4. XGBoost Pairwise Fusion

The final salamander branch treats clustering as pairwise edge selection. We train an XGBoost classifier [16] on labelled salamander pairs and apply it to candidate test pairs. Features include route similarities, supervised-student similarity, frozen descriptor fusion similarity, ORB local evidence, and later yellow-pattern features extracted from SAM-masked views [17]. Candidate pairs whose predicted score passes the selected threshold become merge edges; connected components and graph-constraint overlays then form final clusters.

5.5. Curated Constraint Compilation

The pair verification output is compiled as graph constraints rather than directly edited labels. Verified positive pairs become high-priority must-link edges. Verified negative pairs become cannot-link constraints. Within a candidate cluster, edges are considered in descending priority: verified positive edges first, then model-scored edges whose XGBoost probability exceeds the graph threshold. A union operation is allowed only if it does not place a cannot-link pair in the same component.

Algorithm 1 summarizes the graph compiler. The procedure treats verified and model-scored pair evidence as ordered graph edges. Its only non-standard operation is COMPATIBLE: before a merge is committed, the algorithm checks a dynamic forbidden-neighbour map so that cannot-link constraints are propagated after every component merge.

Algorithm 1. Constrained pair-graph compilation

Require base partition C_0; scored candidate pairs $P = \{(i, j, p_{ij})\}$; verified must-links M; verified cannot-links N; graph threshold τ.
Ensure final partition C.

$U \leftarrow$ disjoint-set over all test images.
For every cluster $S \in C_0$, union all images in S in U.
For each current component A, initialise $F[A] \leftarrow \emptyset$. For every $(a, b) \in N$, add $\mathrm{find}_U(b)$ to $F[\mathrm{find}_U(a)]$ and add $\mathrm{find}_U(a)$ to $F[\mathrm{find}_U(b)]$.
$E_M \leftarrow [(i, j, 1) : (i, j) \in M]$ and $E_P \leftarrow \mathrm{sort}_{\mathrm{desc}}\{(i, j, p_{ij}) \in P : p_{ij} \geq \tau\}$ by score p_{ij}; set $E \leftarrow E_M \,\|\, E_P$.
for each $(i, j, s) \in E$ **do**
  $A \leftarrow \mathrm{find}_U(i)$, $B \leftarrow \mathrm{find}_U(j)$.
  if $A = B$ **then** continue.
  if COMPATIBLE(A, B, F) **then**
    $R \leftarrow \mathrm{union}(A, B)$; set $F[R] \leftarrow (F[A] \cup F[B]) \setminus \{A, B\}$.
    Replace every occurrence of A or B in the remaining forbidden sets by R.
  else discard (i, j, s) as a conflicting edge.
end for
$C \leftarrow$ connected components induced by U.
Apply deterministic split and anchor-attachment overlays, and return C.

This makes the constraint layer a controlled repair stage rather than an unbounded fragmentation step. Compiler thresholds and overlay counts are moved to Appendix A.

6. Species-Specific Routes

6.1. LynxID2025

Lynx images have stable coat patterns and a relatively small number of identities compared with the test image count. We therefore used a supervised MegaDescriptor route as the base, then applied transductive seed smoothing. Stable high-confidence seed clusters define centroids, and embeddings close to a seed centroid are moved toward that centroid before threshold clustering. On local validation, seed smoothing improved ARI from 0.4937 to 0.5625. A single-factor official submission also improved the public leaderboard from 0.48524 to 0.48670, with a later alpha setting reaching 0.48755. Additional co-association probes are summarized in Appendix B.

6.2. SalamanderID2025

Salamander was the most engineering-heavy branch because global descriptors alone did not reliably resolve local yellow-pattern ambiguity. The branch therefore converted retrieval into pair evidence: descriptors and ORB local matching supplied high-recall candidate pairs, while XGBoost and graph constraints selected which local evidence should become cluster edges. After the Texas horned lizard branch showed that local top-K retrieval plus pair-constraint verification could produce high-value trusted evidence, we reused the same workflow for salamanders. The visible identity cue changes from ventral lizard spots to dorsal yellow spot patterns, so the final branch combined frozen descriptor fusion, a supervised MiewID-based student, ORB local evidence, XGBoost pairwise fusion, yellow-pattern features, and graph constraints. ORB itself detects grayscale keypoints and does not know which structures are yellow; the yellow route therefore first extracts a colour-based region of interest and then restricts or summarizes local matching

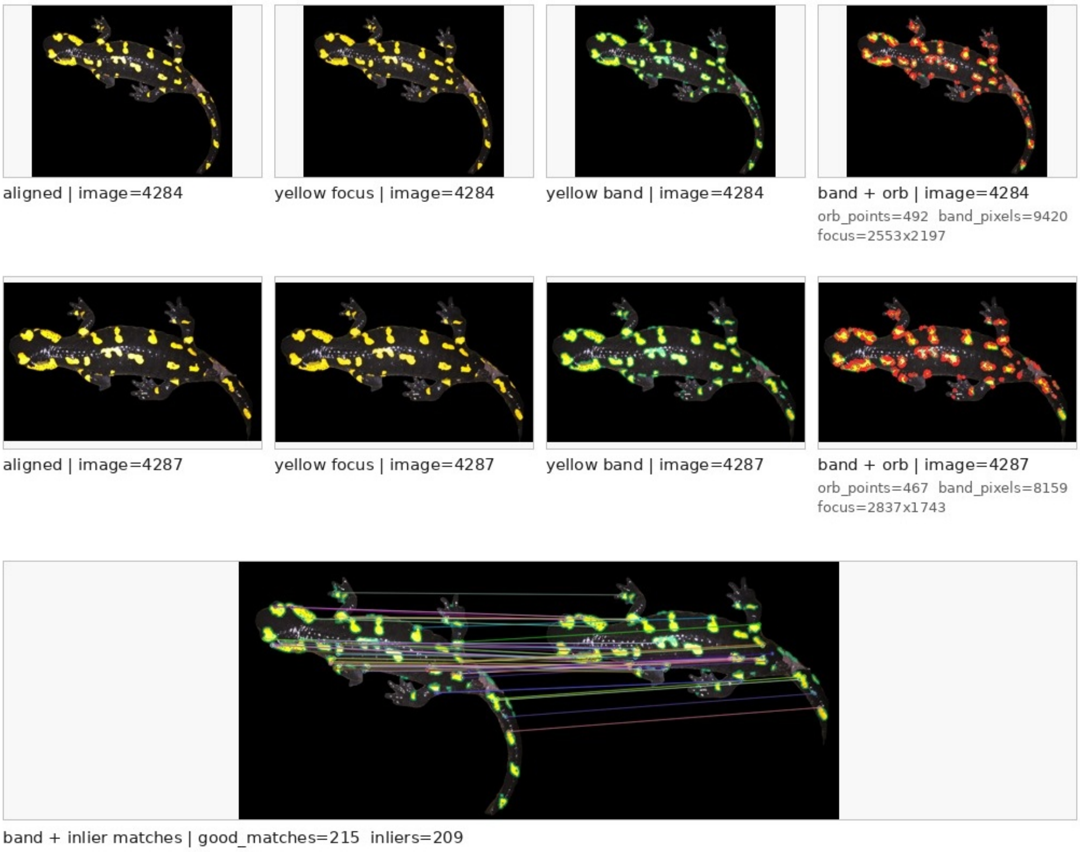

Figure 3: Yellow-pattern guided local evidence for a salamander candidate pair. The upper rows show aligned views, yellow-focused views, yellow-band views, and ORB keypoints. The bottom panel shows inlier matches after local geometric verification. This illustrates why yellow-pattern information is useful as pairwise evidence but should be consumed by a learned or verified graph stage rather than as a hard global veto.

evidence around the visible yellow markings. Figure 3 illustrates why yellow-pattern information should guide candidate scoring rather than act as a hard global veto; the implementation stages and negative cue probe are listed in Appendix B.

The final competition gains came from scaling the same idea to higher-recall salamander candidates. The automatic models supplied candidate recall, while pair-constraint verification supplied enough precision to repair the clustering graph through targeted pair evidence.

6.3. SeaTurtleID2022

Sea turtle was the most stable branch. Frozen MegaDescriptor+MiewID fusion already reached 0.9256 validation ARI, and supervised routes did not reliably improve it. We therefore kept this branch conservative, adding only a cropped-view fusion branch that improved the public leaderboard from 0.49605 to 0.49726 when applied on top of the current mixed base.

6.4. TexasHornedLizards

`TexasHornedLizards` was released without labelled train images, so the original descriptor fallback was poorly calibrated. Lower-threshold descriptor variants helped, but the first major recovery came from a Texas-specific self-training route, which increased the public leaderboard to 0.45751.

The official Texas horned lizard reference paper by Biffi et al. [12] confirmed that individual identity can be read from ventral spot patterns. In their HotSpotter experiment, the first-ranked

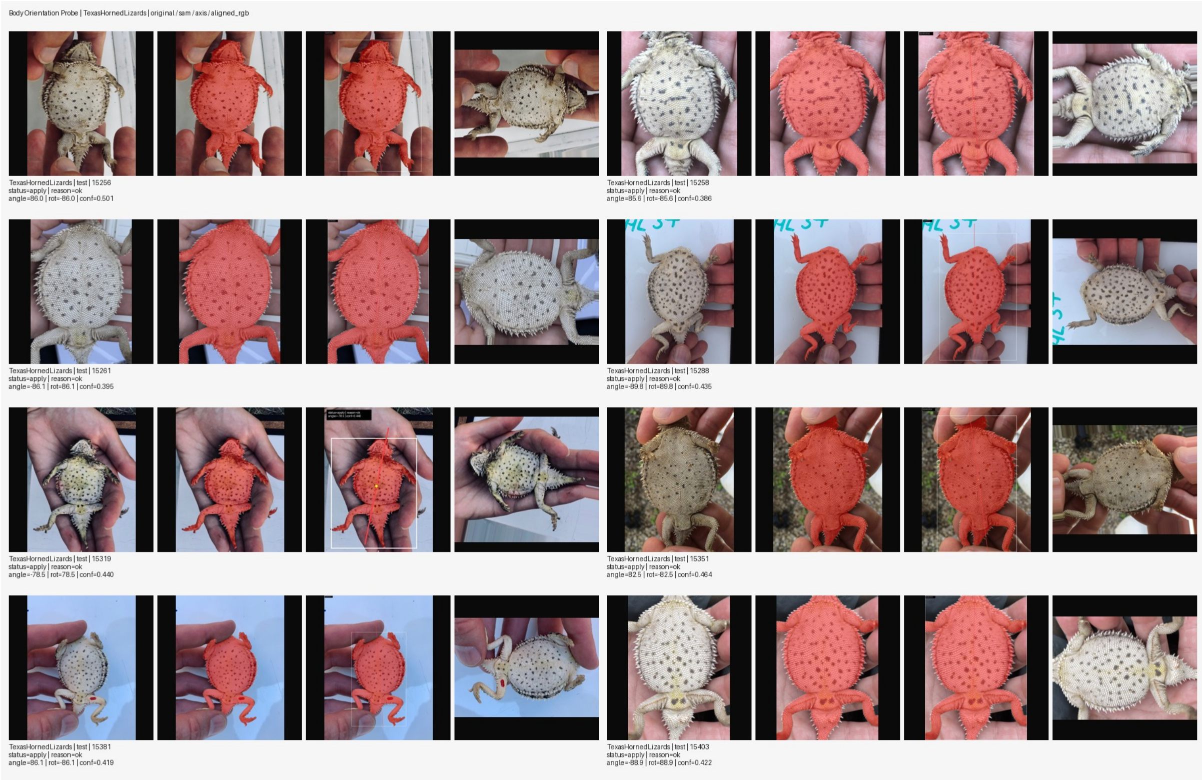

Figure 4: Early Texas horned lizard preprocessing audit. Each group compares the original image, segmentation mask, estimated body axis, and aligned body view. The audit showed that ventral spot layouts could be made more comparable by masking and orientation normalisation before local pair matching.

candidate was correct for 94.4% of known recapture photos, while most remaining failures still contained the true match in the top-10 list. This directly shaped our route: use SAM-aligned [17], gray-percentile, center-body-square views to expose ventral markings, retrieve inclusive top-K candidates, and then compile trusted candidate links into the clustering graph. Figure 4 shows the early data-analysis audit that led to this view-normalisation route.

The self-training route used pseudo-labels only after view normalization and conservative candidate filtering. We first constructed SAM-aligned gray center-body views, retrieved top-K neighbours using descriptor and local evidence, and accepted only high-confidence links as trusted pseudo-positive pairs. These pairs, together with single-view consistency positives from all images, produced a small trusted training set for a Texas-specific MiewID student. The remaining bottleneck was graph connectivity: the route needed more reliable positive connections than descriptor thresholding alone could provide. Figure 5 shows the kind of local ventral-pattern evidence used by this interface.

The final Texas route trained a MiewID-based student using trusted supervision plus all-image single-view pseudo-positive consistency. Replacing only the Texas branch with this trusted-view route increased the public leaderboard from 0.51595 to 0.53646. More importantly, Texas established the pair-constraint principle later reused for salamanders: retrieve high-recall candidate pairs automatically, verify only ambiguous pairs, and compile the trusted decisions into a stronger clustering graph. Additional construction details are given in Appendix B.

7. Results and Submission Trace

The official submission history mixes local validation evidence, public leaderboard feedback, private leaderboard scores released after the competition, and post-hoc references. We therefore mark the source of each score in Table 4. The table is an engineering trace rather than a

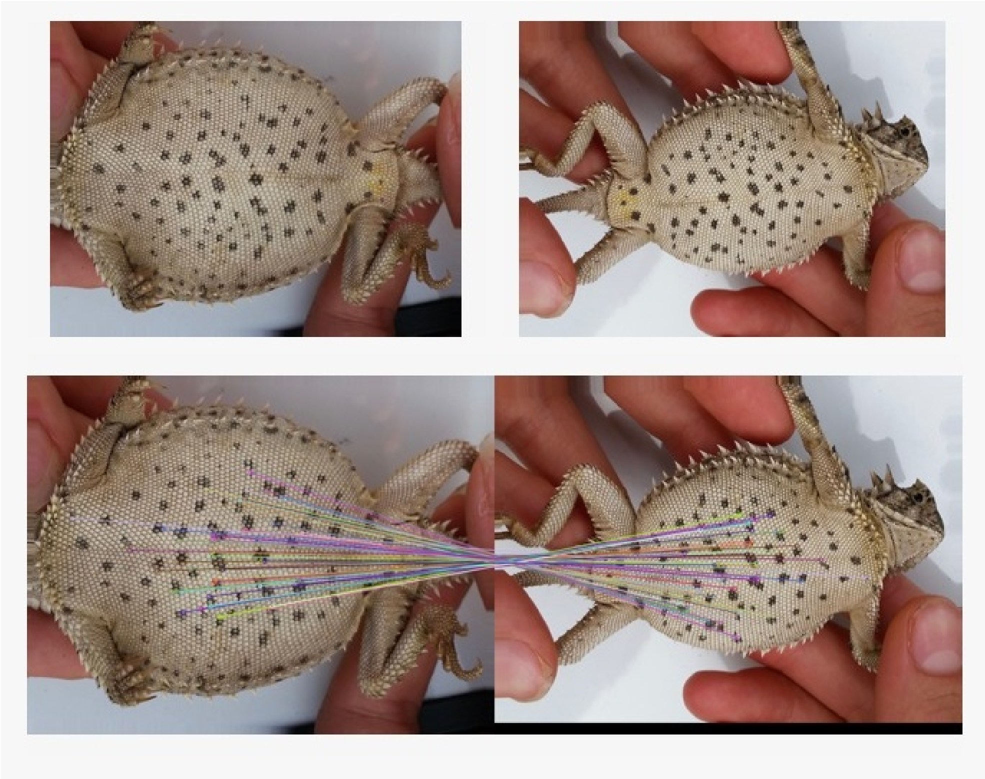

Figure 5: Local ventral-pattern evidence for a Texas horned lizard candidate pair. The top row shows two candidate images after view normalization; the bottom row overlays ORB inlier matches on ventral spot patterns. The example motivates using local matching as a high-recall candidate generator followed by pair-constraint verification and trusted graph compilation.

controlled factorial ablation.

Before manual pair review, the public-best routed base reached 0.51595 public ARI. Later route updates produced a pre-final base with public/private ARI 0.62817/0.65096. The final salamander pair constraints then raised the score to 0.67987/0.69632. This should be read as submission-level evidence, not as a controlled single-factor ablation. Our final submission was uploaded on 2026-04-29 03:04:38 UTC under team name VIPL-VSU and ranked third on the private leaderboard.

Additional cluster-shape diagnostics and negative probes are reported in Appendix C.

8. Discussion

The main observation is that AnimalCLEF 2026 is not only a representation-learning task. It is a partitioning problem under species-specific evidence, long-tailed identities, open-set test clusters, and incomplete local supervision. We found that strong pretrained descriptors were necessary but not sufficient. The most valuable work happened after candidate retrieval: deciding which pair evidence was trustworthy, which dataset should receive local matching, how to calibrate thresholds, and how to keep local validation evidence separate from official leaderboard observations.

Another lesson is that a test-only clustering graph can have an insufficient information budget for some individuals. A single weak view may not expose enough distinctive markings to decide a merge or split, so the useful route is often to expose better identity evidence before clustering. Sea turtle benefited from cropped-view fusion, Texas horned lizards from SAM-aligned body views, and salamanders from local yellow-pattern pair evidence.

The pair verification component should be interpreted narrowly. It was restricted to retrieved

Table 4
Submission trace by intervention stage. Scores are ARI. The table mixes official leaderboard rows and post-hoc references; the source column indicates how each number should be interpreted.

Stage	Source	Public ARI	Private ARI	Main change
Data-routed base	public LB	0.37277	–	Lynx MegaDescriptor fine-tuning; salamander Miew/Mega fusion plus ORB-enhanced clustering; sea turtle frozen fusion; Texas unsupervised descriptor clustering.
Texas self-training	public LB	0.45751	–	SAM-aligned Texas views, high-confidence top-K pseudo-positive links, and a Texas-specific MiewID student.
Salamander XGBoost fusion	public LB	0.48524	–	Learned pairwise fusion over global similarities, branch-score differences, ORB score, inliers, and match ratios.
Yellow-ROI local evidence	public LB	0.51595	0.44839	Colour-based yellow ROIs focused salamander local evidence before ORB-based matching features were fused.
Texas pair registry	public/private LB	0.62528	0.62091	Reviewed Texas pair registry used to repair a small number of cluster errors.
Pre-final routed base	public/private LB	0.62817	0.65096	Last routed base before adding the final salamander pair constraints.
Salamander pair constraints	public/private LB	0.67578	0.69632	Reviewed top-K salamander candidate pairs compiled into a graph overlay.
Final submission	final public/private LB	0.67987	0.69632	Additional salamander cross-view positive pairs; final submission by VIPL-VSU, private rank 3.

candidate pairs and ambiguous clusters, then compiled into deterministic graph operations. The decisions were pair-level constraints over visible image evidence, not hidden labels or direct identity assignments. This made the constraints inspectable, but it also means that the final constrained submission is human-in-the-loop. We therefore separate the automatic routed system from the final constrained submission in Table 4.

The largest limitation of this note is that it records a competition pipeline whose final branch selection depended on public leaderboard feedback and many local artifacts. We therefore report leaderboard gains mainly as intervention-level evidence rather than as a controlled factorial ablation. For reproducibility, this note reports the branch-level preprocessing, model choices, thresholds, pair-review counts, and graph-compilation settings used by the submitted pipeline. The final submitted CSV was uploaded on 2026-04-29 03:04:38 UTC and is associated with the public/private scores reported in Table 4.

9. Conclusion

We presented a dataset-routed hybrid clustering pipeline for AnimalCLEF 2026 that obtained 0.67987 public ARI and 0.69632 private ARI. The central lesson is that this benchmark was not solved by a single embedding or threshold. Performance improved when each dataset exposed its own identity evidence and when uncertain retrieval links were converted into graph constraints. For lynx and sea turtles, conservative descriptor clustering was sufficient; for salamanders and Texas horned lizards, local candidate-pair evidence and constraint compilation were the decisive parts of the final pipeline.

Code and Artifact Availability

We prepared a public release for the VIPL-VSU AnimalCLEF 2026 working note at https://github.com/geyaokai/animalclef26. The release includes the reusable pipeline code, runnable scripts, smoke tests, the final submitted CSV, compact salamander and Texas pair-constraint artifacts, route summaries, and selected checkpoints that fit GitHub LFS limits. Two checkpoints are tracked with Git LFS. The MegaDescriptor-based lynx checkpoint is larger than the GitHub LFS per-object limit, so its configuration and reports are included and the checkpoint should be distributed through a separate artifact host. Raw images, large caches, generated review boards, and private local staging/audit artifacts are not included.

Acknowledgments

We thank the AnimalCLEF 2026 organizers for preparing the benchmark, maintaining the Kaggle competition, and providing the evaluation infrastructure. We also thank the maintainers and authors of the open-source animal re-identification resources and pretrained models used in this work, including MiewID, MegaDescriptor, WildlifeDatasets, scikit-learn, OpenCV, and XGBoost.

Declaration on Generative AI

During the preparation of this work, the authors used generative AI tools for coding assistance, experiment organization, and language editing. The method design, experiments, result interpretation, and final manuscript content were reviewed and are the responsibility of the authors.

A. Implementation Details

MegaDescriptor-L takes 384×384 RGB inputs and outputs a 1536-dimensional vector. MiewID-msv3 takes 440×440 RGB inputs and outputs a 2152-dimensional vector. Early fusion concatenates the two normalised blocks into a 3688-dimensional vector and normalises the result again. For supervised routes, each student backbone is followed by a linear projection to a

Table 5
Main branch configuration for the final submitted pipeline.

Dataset	Route	Dim.	Threshold
`LynxID2025`	MegaDescriptor-L supervised student with seed smoothing	512	0.85
`SalamanderID2025`	Miew masked-SupCon student + frozen fusion + ORB/XGBoost/yellow features + graph constraints	4200	0.25
`SeaTurtleID2022`	frozen MegaDescriptor+MiewID fusion with cropped fusion branch, weight 0.3	7376	0.7
`TexasHornedLizards`	Miew trusted-view self-training on SAM-aligned gray center-body views	512	0.44

Table 6
Key operating parameters of the submitted pipeline. Thresholds are cosine-distance clustering thresholds unless noted otherwise.

Dataset	Main evidence	Gate	Pair-constraint role
`LynxID2025`	MegaDescriptor supervised student, TTA variants, transductive seed smoothing	0.85	not used; descriptor graph was sufficient
`SalamanderID2025`	Miew supervised student, frozen fusion, ORB top-$K = 10$, yellow-pattern features, XGBoost pair fusion	graph 0.25; ORB inliers $\geq$8	positive/negative constraints for ambiguous candidates
`SeaTurtleID2022`	frozen MegaDescriptor+MiewID fusion plus cropped-view fusion with weight 0.3	0.7	not used; conservative descriptor fusion was selected
`TexasHornedLizards`	SAM-aligned gray center-body views, Miew trusted-view self-training, ORB candidate evidence	0.44	automatic trusted pairs first; later pair review for ambiguous candidates

512-dimensional L2-normalised embedding. Training uses separate ArcFace heads [18] for each dataset rather than one shared closed-set head. We also evaluated feature distillation, relation distillation, masked supervised contrastive loss [19], and a salamander sub-center ArcFace head [18].

ORB extraction used up to 1024 features per image after resizing the longer side to 768 pixels. Candidate matching was implemented with OpenCV [20] using brute-force descriptor matching, a ratio test of 0.8, and RANSAC geometric verification [15]; pairs with fewer than 8 inliers received zero local support. The local score was used as a conservative boost:

$$s_{\text{rerank}} = s_{\text{global}} + w_{\text{local}} s_{\text{local}} (1 - s_{\text{global}}). \tag{2}$$

Here s_{global} is the descriptor cosine similarity, s_{local} is the normalized ORB/RANSAC local evidence, and w_{local} controls how strongly local matches can boost a pair; the factor $(1 - s_{\text{global}})$ keeps the boost small for pairs that are already globally similar. In the salamander smoke audit, positive pairs had mean local score 0.3366, while negative pairs had mean local score 0.0111.

For the gate2 salamander compiler, we used `graph_threshold=0.25`, `min_judged_pairs=2`, and `min_no_pairs=2`. The gate compiled 42 candidates, skipped 61 low-evidence candidates, changed 89 salamander images, and produced 60 overlay operations. Compared with a split-only base, it reduced salamander clusters from 427 to 412 and singletons from 260 to 229.

Table 7
Manual pair-constraint artifacts used in the final constrained submission.

Dataset / stage	Reviewed pairs	Changed images	Effect
Salamander pair constraints	179 yes / 4216 no or default no	200	top-K candidate-pair review; public/private ARI 0.62817/0.65096 $\rightarrow$ 0.67987/0.69632
Texas pair registry	86 yes / 1514 no	8	small graph repair on ambiguous Texas clusters

B. Additional Route Details

For lynx, we also tested route co-association over multiple supervised routes and thresholds. The useful variants combined synth-warmup and previous-best TTA routes through average, mutual-kNN, and complete-link style predictions, then kept pairs supported by a high co-association vote threshold. These probes were secondary to the final pipeline because the lynx branch was already reasonably stable under supervised metric learning and seed smoothing.

For salamanders, the final route was built in five stages: frozen MegaDescriptor+MiewID fusion; ORB local reranking only for salamanders; a MiewID-based supervised embedding trained with masked supervised contrastive loss and concatenated with frozen fusion features; XGBoost pairwise fusion over global, local, and route-specific features; and split or pair-constraint judgments compiled into a graph overlay. A yellow-only veto looked plausible because this dataset is dominated by yellow dorsal markings, but it over-fragmented clusters and reduced local validation ARI from 0.5875 to 0.2905 in one probe. The same cue, when introduced as an XGBoost feature, improved the public leaderboard from 0.49726 to 0.51082.

For Texas horned lizards, the trusted-view student used 46 trusted images and 228 untrusted images, producing 1142 training rows after view construction. The final selected route used trusted supervision plus all-image single-view pseudo-positive consistency, without teacher distillation. The branch was valuable not only for its own leaderboard gain but also because it established the top-K candidate retrieval and pair-verification interface later reused on salamanders.

C. Additional Results and Probes

Table 8
Cluster shape of the final constrained submission.

Dataset	Samples	Clusters	Singleton clusters	Singleton ratio
`LynxID2025`	946	21	2	0.095
`SalamanderID2025`	689	551	482	0.875
`SeaTurtleID2022`	500	153	41	0.268
`TexasHornedLizards`	274	213	175	0.822

Several negative results shaped the final pipeline. Descriptor concatenation for Texas preserved high-confidence seed pairs but degraded the public score, suggesting that the proxy did not penalise outsider merges near seed clusters. Salamander yellow cues were harmful as hard vetoes but helpful as learned pairwise features. Deep local matching probes with ALIKED [21] and LightGlue [11], plus a WildFusion-style calibration, did not outperform the simpler ORB-based route in our validation protocol. Dual-view SAM-masked salamander routes were engineering-ready but did not improve official submissions when directly inserted into the current XGBoost pipeline.

References

[1] S. Schneider, G. W. Taylor, S. Linquist, S. C. Kremer, Past, present and future approaches using computer vision for animal re-identification from camera trap data, Methods in Ecology and Evolution 10 (2019) 461–470.

[2] V. Cermak, L. Picek, L. Adam, K. Papafitsoros, Wildlifedatasets: An open-source toolkit for animal re-identification, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 5953–5963.

[3] L. Picek, L. Adam, S. Kahl, R. Bossy, L. Chrobak, H. Goëau, K. Papafitsoros, H. Klinck, W.-P. Vellinga, R. Planqué, T. Denton, K. Barnard, C. Nédellec, L. Deléger, M. Courtin, G. Martellucci, I. Moummad, F. Vinatier, P. Bonnet, A. Joly, Overview of LifeCLEF 2026: Ai challenges for biodiversity understanding and ecosystem management, in: International Conference of the Cross-Language Evaluation Forum for European Languages (CLEF), Springer, 2026.

[4] L. Adam, K. Papafitsoros, D. A. Williams, D. Biffi, L. Picek, Overview of AnimalCLEF 2026: Discovery and re-identification of individual animals, in: Working Notes of CLEF 2026 - Conference and Labs of the Evaluation Forum, 2026.

[5] L. Hubert, P. Arabie, Comparing partitions, Journal of Classification 2 (1985) 193–218.

[6] D. Sani, M. Khurana, S. Anand, Active learning for animal re-identification with ambiguity-aware sampling, 2025. ArXiv:2511.06658.

[7] Conservation X Labs, Miewid: A multi-species animal re-identification model, 2024. ArXiv:2412.05602.

[8] L. Adam, V. Čermák, K. Papafitsoros, L. Picek, SeaTurtleID2022: A long-span dataset for reliable sea turtle re-identification, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops, 2024.

[9] L. Adam, V. Čermák, L. Picek, et al., CzechLynx: A dataset for individual identification and pose estimation of the eurasian lynx, Scientific Data (2026).

[10] E. Rublee, V. Rabaud, K. Konolige, G. Bradski, ORB: An efficient alternative to SIFT or SURF, in: Proceedings of the IEEE International Conference on Computer Vision, 2011, pp. 2564–2571.

[11] P. Lindenberger, P.-E. Sarlin, M. Pollefeys, LightGlue: Local feature matching at light speed, 2023. ArXiv:2306.13643.

[12] D. Biffi, M. R. Tucker, A. Ackel, D. A. Williams, Identification of individual texas horned lizards (Phrynosoma cornutum) using genotypes and ventral spot patterns, Ecology and Evolution 15 (2025) e71167.

[13] L. Adam, K. Papafitsoros, L. Picek, et al., Overview of AnimalCLEF 2025: Recognition of individual animals in images, in: Working Notes of CLEF 2025 - Conference and Labs of the Evaluation Forum, 2025.

[14] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay, Scikit-learn: Machine learning in python, Journal of Machine Learning Research 12 (2011) 2825–2830.

[15] M. A. Fischler, R. C. Bolles, Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography, Communications of the ACM 24 (1981) 381–395.

[16] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794.

[17] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, R. Girshick, Segment anything, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 4015–4026.

[18] J. Deng, J. Guo, N. Xue, S. Zafeiriou, ArcFace: Additive angular margin loss for deep face

recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 4690–4699.

[19] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, D. Krishnan, Supervised contrastive learning, in: Advances in Neural Information Processing Systems, volume 33, 2020, pp. 18661–18673.

[20] G. Bradski, OpenCV: Open source computer vision library, Dr. Dobb's Journal of Software Tools (2000).

[21] X. Zhao, X. Wu, W. Chen, P. C. Y. Chen, Q. Xu, Z. Li, ALIKED: A lighter keypoint and descriptor extraction network via deformable transformation, IEEE Transactions on Instrumentation and Measurement 72 (2023) 1–16.